



Nikoo Asadnejad

Software Engineer

Senior Software Engineer with more than 5 years of experience designing and developing scalable, high-performance applications using **C#**, **.NET**, and modern software architecture principles in Agile/Scrum environments.

Experienced in building distributed systems, microservices, and business-critical platforms using **DDD**, **CQRS**, **gRPC**, **RabbitMQ**, **Redis**, **MongoDB**, **SQL Server**, and **REST APIs**. Strong background in System Design, **Software Architecture**, **Performance Optimization**, and **Observability**, with a focus on reliability, maintainability, scalability, and business impact.

Beyond the Microsoft ecosystem, I have foundational knowledge of **Python** and **Go**, and I'm actively exploring AI Engineering, **Agentic AI**, and cloud-native technologies through personal projects and continuous learning. I also leverage AI-powered development tools, including **Claude Code** and **Cursor**, to enhance productivity and accelerate software delivery.

Currently pursuing a **Master's** degree in Software Engineering, with a strong interest in **AI research**.

Passionate about **solving** complex engineering **challenges**, collaborating with talented teams, continuously learning new technologies, and building innovative solutions that deliver meaningful business value.

Contact

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Medium:

<https://medium.com/@nikoo.asadnejad.work>

Technical Skills

- Software Engineering
- Software Architecture
- System Design
- MicroServices, Distributed Systems
- Clean Architecture,DDD, CQRS, Event-Sourcing
- Event-Driven, RabbitMq
- Design Patterns, SOLID, OOP
- C#.Net, SQL Server, EF
- MongoDb,Redis,ElasticSearch
- Restful Api, gRpc, SignalR, MVC, WCF, GraphQL
- GIT, CI/CD
- Unit Testing , xUnit, TDD
- Linux, Docker & Kubernetes
- Python & Bash Scripting
- HTML & CSS
- JavaScript & JQuery

Language

English:Professional French:Basic

Experience

○ Oct2022 - Present | Fly Today

Senior Back-End Engineer

Responsibilities and Achievements:

- Took end-to-end **ownership** of *Train, Bus, Activities Booking* systems and *Hotel Recommendation* systems, from **requirements analysis** and **architecture design** through *development, deployment, monitoring*, and *maintenance*.
- Designed and developed scalable systems using **Clean Architecture**, **DDD**, **CQRS**, **Microservices**, **SOA**, and **Modular Monoliths**, with **gRPC**, **RabbitMQ**, **SignalR**, **YARP**, and **Ocelot**.
- Designed a Financial **Ledger** system using **Event Sourcing**, focusing on transaction traceability, consistency, and auditability.
- **Improved API throughput and performance** through **Redis** and in-memory **caching**, **async/parallel programming**, **SQL query optimization**, **MongoDb** indexing.
- Implemented an external Wallet system using **Chain of Responsibility**, **Strategy**, and **Factory** design patterns, focusing on extensibility and maintainability.
- Implemented **fuzzy** and **semantic** hotel search using **Elasticsearch** to improve search relevance.
- Improved *observability* and *reliability* using **OpenTelemetry**, **ELK**, **Prometheus** distributed **tracing**, **metrics**, **health checks**, **retry policies**, **circuit breaker**.
- Designed **CI/CD** pipelines, containerized applications with **Docker**, and developed **unit and architectural tests** to ensure code quality and architectural consistency.
- Conducted **code reviews**, **mentored** junior developers.

○ Aug 2021 - Oct 2022 | Kahkeshan Group of Companies | Karan

Back-End Engineer

Responsibilities and Achievements:

- Developed new features and refactored legacy APIs in the Karan project
- (**12 microservices**), using ASP.NET Core with **SQL Server**, **Redis**, and **MongoDB**.
- Deployed and maintained a web crawler application and Telegram bots on **Linux** servers.
- **Optimized** search queries, achieving **3x faster** performance.
- Designed and implemented a multi-gateway payment service using the **Factory Pattern** for flexible payment integrations.
- Built a JobAlert matching service with **RabbitMQ** and **Quartz jobs** to automatically pair resumes with relevant job ads based on skills and requirements.

○ Sep 2018 - Oct 2022 | Kahkeshan Group of Companies | Karan
Software Support and QA Specialist

Education

2024-now

Master / Software Engineering | Azad University, Science And Research Branch
2018-2022

Bachelor / Software Engineering | Azad University Tehran North Branch

Licenses & Certifications

- **Docker And Kubernetes**

CanDo Docker , Docker Swarm , Kubernetes
Issued Feb 2025 60h

- **LIPC-2** 50h

Arjang Linux Kernel , Systemd , NGINX ,
Issued Jul 2024

- **LIPC-1** 45h

CanDo | Linux Commands, Bash Scripts
Issued Apr 2024

- **Asp.Net Core**

Kahkeshan Noor 20h 140h
Issued Nov 2021 | .NET Core , Rest API

- **MCSA Web Applications**

Kahkeshan Noor
Issued Aug 2020 | C# , .NET , ASP.NET , MVC

- **Web Design**

Tehran Institute Of Technology 80h
Issued Apr 2019 | HTML , CSS , JavaScript , JQuery

Projects

- **NotificationService**

<https://github.com/Nikoo-Asadnejad/NotificationService>

A microservice-based notification platform built with asynchronous communication using **RabbitMQ**. It provides a central API for processing notification requests and dispatching them to dedicated Email, SMS, and Push Notification microservices, enabling scalable, reliable, and decoupled message delivery.

- **Linkedin Post Creator Agent**

https://github.com/Nikoo-Asadnejad/post_creator_agent

AI LinkedIn Content Generator – A multi-agent AI system built with **Python**, **LangChain** (LCEL), and Ollama that automatically generates ready-to-publish LinkedIn posts and accompanying images. The workflow uses asynchronous parallel execution, where a search agent researches topics when needed, while content and image generation agents run concurrently to produce accurate, fact-based content with minimal hallucinations.

- **DevAgent**

<https://github.com/Nikoo-Asadnejad/DevAgent>

DevAgent turns a GitHub issue into a draft pull request. Built with **Python**, **FastAPI**, **LangGraph**, **Pydantic**, **OpenHands**, Docker, and Gitleaks, it clones the repository and creates an isolated sandbox for each task. OpenHands uses the Agent Client Protocol (**ACP**) to run a local coding agent—currently Codex—inside that sandbox. DevAgent then builds, tests, and scans the changes before opening a draft PR for review.

- **ManufacturingSystem**

<https://github.com/Nikoo-Asadnejad/ManufacturingSystem>

A .NET 10 worker that models a manufacturing workflow driven by temperature and pressure measurements. At each polling interval, it combines the latest measurements into an immutable sensor snapshot, evaluates workflow rules, and runs eligible stages while coordinating shared machine resources. For asynchronous processing, a **BroadcastBlock<T>** distributes sensor measurements to separate, bounded **BufferBlock<T>** consumers for temperature and pressure. A bounded .NET **Channel<T>** carries completed snapshots to the workflow consumer, which dispatches each snapshot to the registered workflows. Resource access uses **SemaphoreSlim** to prevent **race conditions**, **Interlocked** and **Volatile** for atomic state changes and memory visibility, and consistent resource ordering to prevent deadlocks.

- **Kibana Dashboard Generator**

https://github.com/Nikoo-Asadnejad/kibana_dashboard_generator

This project provides a fully automated solution for generating **Kibana** dashboards directly from a Swagger/OpenAPI specification. The script analyzes each controller (API scope) defined in the specification and produces a dedicated dashboard containing detailed monitoring panels for every endpoint. Each panel visualizes the API's operational metrics, including the distribution of returned HTTP status codes, enabling teams to observe performance patterns, detect anomalies, and quickly identify reliability issues.